# EMCH 367 / ELCT 331

## Modern Control Systems for Physical Engineering Applications

University of South Carolina

**Fall 2026**

| | |
|---|---|
| **Cross-listed:** EMCH 367 / ELCT 331 | **Instructor:** J.C. Vaught |
| **Meeting Time:** Tue/Thu 10:05–11:20 AM | **Office:** TBD |
| **Location:** TBD | **Office Hours:** TBD |

## Catalog Description

This course develops practical competency in modeling, feedback control, estimation, and implementation for real engineered systems. The course is concept-first in lecture and application-heavy in assignments. Students learn to build control-ready models of physical systems, design and evaluate feedback controllers, implement controllers digitally, handle key nonlinear effects, and work with multivariable systems using state space methods. The course includes two team projects with physical demonstrations and presentations.

## Prerequisites

Students are expected to be fluent in calculus through multivariable calculus, differential equations, and linear algebra. Students must be comfortable writing and debugging code in Python or MATLAB and producing plots and numerical summaries. **The course does not include a formal review of math prerequisites.**

## Required Resources

Students need a **laptop** capable of running the course programming environment. The course will standardize on a primary environment (Python or MATLAB) and provide a reference environment for grading. Students must be able to run provided starter code, produce required plots, and submit reproducible results. Hardware resources and safety requirements for demos are provided by the course and described on the course site.

## Instructional Approach

Lecture emphasizes **concepts, design reasoning, and engineering workflows** rather than step-by-step derivations. Mathematical derivations and deeper proofs are provided through optional typed notes and video resources. **Theory homework** requires formal engagement with selected mathematical tools, while **application homework** emphasizes implementation, validation, and interpretation.

## Assessment Weights

**This course is not curved.** Final grades are determined by the following components.

| Component | Weight |
|---|---|
| **Homework (11 sets, lowest dropped)** | 50% |
| **Projects** | 40% |
| **Project 1** | 15% |
| **Project 2** | 25% |
| **Exams (two in-term, 5% each)** | 10% |

## Grading Scale

Final letter grades are assigned according to the following scale.

| A | B+ | B | C+ | C | D+ | D | F |
|---|---|---|---|---|---|---|---|
| 90–100 | 85–89 | 80–84 | 75–79 | 70–74 | 65–69 | 60–64 | <60 |

## Homework

There are **11 homework sets** total. The **lowest score is dropped**. The remaining 10 are weighted by rank, with the lowest grade receiving the highest weight.

| **Rank** | 1st | 2nd | 3rd | 4th | 5th | 6th | 7th | 8th | 9th–10th |
|---|---|---|---|---|---|---|---|---|---|
| **Weight** | 10% | 8% | 6% | 6% | 4% | 4% | 3% | 3% | 3% each |

**HW0** is a **mathematics prerequisites assessment** due **Sunday at midnight** before the first day of class. This diagnostic helps students identify gaps in their mathematical preparation. **HW1–HW10** each consist of two files: $\mathbf{HW\_X_m}$ (mathematics/theory portion) and $\mathbf{HW\_X_c}$ (computational/application portion).

## Submission Requirements

HW0 is due **Sunday at midnight** before the first class meeting. HW1–HW10 are due **Monday at midnight** before that week's classes. **Typed submissions only**—handwritten work is not accepted; at minimum, all final answers must be typeset (e.g., LaTeX, Word with equation editor, or equivalent). Coding portions are graded on **final output matching spec exactly**, and all code submissions must be **reproducible**.

## Exams

Two in-term exams (5% each) focus on **applied reasoning**: selecting modeling assumptions, choosing control structure, interpreting behavior, specifying engineering steps, and diagnosing failures. Each exam includes an **optional challenge section** worth up to 10% bonus on that exam.

The final exam is **optional** and covers material from all aspects of the course. Students who take the final exam may choose to apply their score as a **replacement** for either one homework grade or one in-term exam grade—whichever benefits the student most. Alternatively, students may choose not to apply the final exam score at all. The final exam cannot hurt your grade; it can only help.

## Additional Credit

Bonus credit is available on individual homework and projects through “above and beyond” extensions defined by the course. **Optional bonus research projects** may also be approved at the beginning of the semester: a 5-point project is expected to be approximately 2× the difficulty of Project 2, and a 10-point project is expected to be approximately 3–4× the difficulty of Project 2. Bonus research projects must be requested before **2 weeks prior to the final exam**.

## Learning Outcomes

By the end of the course, students will be able to achieve the following objectives.

| # | Outcome | Description |
|---|---|---|
| 1 | Model physical systems | Translate a physical system into a control-ready model with explicit assumptions and validity ranges, including actuator and sensor non-idealities |
| 2 | Design feedback controllers | Design baseline feedback controllers and select appropriate control architectures for real systems |
| 3 | Evaluate performance | Evaluate performance using standard time-domain metrics and interpret stability evidence in both time and frequency perspectives |
| 4 | Implement digitally | Implement controllers in discrete time and anticipate common sim-to-hardware failure modes |
| 5 | Handle nonlinearities | Model and reason about nonlinear effects and robustness |
| 6 | Work with multivariable systems | Represent and reason about multivariable systems in state space and understand the motivation for native multivariable controller designs |
| 7 | Communicate effectively | Communicate engineering decisions and experimental evidence in written and oral form |

## Course Projects

Projects constitute a significant portion of the course grade and provide hands-on experience with control system design, implementation, and validation. Students will complete two projects throughout the semester, progressing from single-input single-output (SISO) systems to more complex multi-input multi-output (MIMO) systems with machine learning integration.

### Team Formation

All projects must be completed in teams of **2–3 students**. **Single-person projects are not permitted** due to insufficient time for individual presentations and the collaborative nature of real-world engineering practice.

Students may self-organize into teams during the first lecture. After the first lecture, any remaining unassigned students will be placed into teams by the instructor. No team changes are permitted until after Project 1 is complete. Requests for changes after Project 1 must be submitted in writing with justification and are subject to instructor approval.

### Project 1: SISO Control Implementation

**Weight: 15% of Final Grade**

### Overview

Project 1 focuses on the design and implementation of a controller for a **single-input single-output (SISO)** system. This project establishes foundational skills in control implementation that will be extended in Project 2.

Systems may be **continuous-time** or **discrete-time**. Implementation must be on **real hardware** or a **detailed simulation** with realistic sensor and actuator models (including noise, quantization, delays, and saturation). Projects are **open-ended**—students are encouraged to propose their own systems.

### Example Systems

The following systems are provided as a starting menu. Students may select from this list or propose an alternative system for approval.

Table 1: Example SISO Systems for Project 1

| System | Input | Output |
|---|---|---|
| Inverted pendulum / cart-pole | Cart force or motor voltage | Pendulum angle |
| Vehicle steering control | Steering angle | Lateral position or heading |
| DC motor position control | Motor voltage | Shaft position |
| DC motor speed control | Motor voltage | Shaft angular velocity |
| Ball and beam | Beam angle | Ball position |
| Thermal system regulation | Heater power | Temperature |
| *Custom approved system* | *TBD* | *TBD* |

## Project 2: MIMO with ML-Based Control

**Weight: 25% of Final Grade**

### Overview

Project 2 advances to **multi-input multi-output (MIMO)** systems and introduces machine learning-based control methods. This project emphasizes the comparison between classical and modern control approaches.

Systems should have **multiple inputs and outputs** (preferably many I/O channels). **Implementation must be discrete-time.** Each project **must include both** an ML-based controller (e.g., reinforcement learning, neural network control, adaptive methods) **and** a classical controller for comparison (e.g., LQR, PID, state feedback). Projects are **open-ended**—students are encouraged to continue/expand their Project 1 system or propose a new system.

### Example Systems

## Available Resources

The following resources are available to all project teams.

## Demonstration Requirements

All projects require demonstration of the implemented control system. A recorded **video demonstration** is **required** and must be uploaded with the project submission. Live demonstrations

Table 2: Example MIMO Systems for Project 2

| **System** | **Inputs** | **Outputs** |
|---|---|---|
| Quadrotor attitude/position | Motor speeds (4) | Roll, pitch, yaw, position (6) |
| Multi-joint robotic arm | Joint torques/voltages | Joint angles, end-effector pose |
| Multi-zone HVAC system | Damper positions, fan speeds | Zone temperatures, humidity |
| Autonomous vehicle | Steering, throttle, braking | Position, velocity, heading |
| *Custom approved system* | *TBD* | *TBD* |

Table 3: Available Project Resources

| **Resource** | **Details** |
|---|---|
| 3D Printers | Available for fabricating custom mechanical parts |
| Computers | Workstations with full course software environment |
| Arduino Microcontrollers | Standard issue for all teams |
| Raspberry Pi | Available upon request with written justification of computational requirements |
| Sensor/Actuator Kits | Standardized kits provided to each team |
| *Additional Options:* | |
| Personal Equipment | Students may purchase additional equipment at their own expense |
| Research Lab Equipment | Students may integrate equipment from their research laboratories with advisor approval |

during presentations are permitted *only* if the equipment is small and portable; large or fragile setups should be demonstrated via video only.

Demonstrations must show **multiple trials** to establish repeatability of performance—a single successful run is insufficient. Demonstrations must also include **failure case analysis**: What conditions cause the controller to fail? What are the stability boundaries? How does the system degrade under disturbances or parameter variations?

### Deliverables

Both projects require the following deliverables. All items are mandatory unless explicitly stated otherwise.

Table 4: Required Project Deliverables

| # | Deliverable | Description |
|---|---|---|
| 1 | Project Proposal Document | Submitted for instructor approval *before* work begins. Must include system description, objectives, methodology, and timeline. |
| 2 | Code Repository | All source code must be submitted. Code must be **reproducible** with clear instructions for execution. |
| 3 | Team Report | Comprehensive technical report covering theory, design, implementation, results, and analysis. |
| 4 | Individual Report | Each team member submits a personal reflection on their contributions and key learnings. |
| 5 | Presentation Slides | Submitted *before* presentation day for review. |
| 6 | Live Presentation | In-class presentation followed by Q&A session with instructor and peers. |
| 7 | Demo Video | Recorded demonstration with narration explaining the system operation and results. |

### Grading Criteria

Projects will be evaluated according to the criteria outlined in Table 5. Project 2 places additional emphasis on the rigor of comparison between classical and ML-based controllers.

**Important:** Project proposals must be approved before work begins.
Late submissions will be penalized according to the course late policy.
Academic integrity violations will result in a zero for the project and referral to the Office of Academic Integrity.

Table 5: Project Grading Criteria

| Criterion | Evaluation Focus |
|---|---|
| Correctness of Implementation | Does the controller achieve the stated objectives? Is the implementation technically sound? |
| Quality of Validation & Testing | Are results validated against theoretical predictions? Is testing comprehensive and systematic? |
| Robustness of Performance | How does the system perform under disturbances, noise, and parameter variations? |
| Clarity of Evidence | Are claims supported by clear data and analysis? Are figures and tables well-constructed? |
| Technical Communication | Quality of written reports, oral presentation, and ability to answer questions. |
| Comparison Rigor (Project 2) | Depth and fairness of comparison between classical and ML-based controllers. Both approaches must be given equal effort and evaluation. |

# Course Policies

## AI Tools Policy

| Context | Policy |
|---|---|
| **Application Assignments** | AI tools **allowed**. Must explain approach, interpret results, and verify correctness. |
| **Theory Assignments** | Submissions must reflect the **student's own reasoning**. |
| **Exams** | **No AI tools permitted.** |
| **Disclosure** | Must disclose AI use in a tool-use note with each submission. |

## Collaboration

Conceptual discussion with classmates is **allowed and encouraged**, but all submissions must be your **own work**. Project work must reflect **equitable contribution** from all team members. External resources must be **cited**.

## Late Work

Late homework submissions are generally not accepted because one lowest score is dropped. Project deadlines are firm due to hardware scheduling and presentation logistics.

## Accessibility

Students with accommodations should coordinate with the instructor **early in the semester** to ensure appropriate arrangements can be made.

## Course Schedule

The course meets Tuesdays and Thursdays throughout Fall 2026. The schedule below maps all 28 meetings to specific dates, accounting for university holidays and breaks.

In the schedule table, shaded rows indicate exam sessions, blue-shaded rows indicate project presentation sessions, and **bold garnet text** indicates homework due dates and project milestones.

**HW0 (Math Prerequisites):** Due **Sunday, August 16** at midnight.

| Wk | Date | Mtg | Type | Topic |
|---|---|---|---|---|
| 1 | Thu 8/20 | 1 | Content | Modeling Physical Systems for Control |
| | | | | **HW1 Due Mon 8/24**: Modeling I |
| 2 | Tue 8/25 | 2 | Content | Modeling Physical Actuators for Control |
| | Thu 8/27 | 3 | Content | Modeling Sensor Error for Control |
| | | | | **HW2 Due Mon 8/31**: Modeling II |
| 3 | Tue 9/1 | 4 | Content | Designing Control Systems |
| | Thu 9/3 | 5 | Content | Measuring Controller Performance |
| | | | | **HW3 Due Tue 9/8**: Basic Feedback |
| 4 | Tue 9/8 | 6 | Content | Stability Analysis in Time and Frequency Domain |
| | Thu 9/10 | 7 | Content | Control Architecture |
| | | | | **HW4 Due Mon 9/14**: Stability |
| 5 | Tue 9/15 | 8 | Content | Introducing State Space Representation |
| | Thu 9/17 | 9 | Buffer | Integration Studio and Exam 1 Prep |
| | | | | **HW5 Due Mon 9/21**: Architecture & State Space |
| 6 | Tue 9/22 | 10 | Exam | **Exam 1**: Modeling, Feedback, Stability |
| | Thu 9/24 | 11 | Content | Implementing Control in Discrete Domain |
| | | | | **Project 1 Proposal Due Thu 9/24** |
| 7 | Tue 9/29 | 12 | Content | Nonlinear System Accounting and Control Robustness |
| | Thu 10/1 | 13 | Content | Estimating State in Nonlinear Systems |
| | | | | **HW6 Due Mon 10/5**: Discrete & Nonlinear |
| 8 | Tue 10/6 | 14 | Presentation | Project 1 Presentations Day 1 |
| | Thu 10/8 | 15 | Presentation | Project 1 Presentations Day 2 |
| | | | | **Project 1 Report Due Thu 10/8** |
| 9 | Tue 10/13 | 16 | Content | Modeling MIMO Systems in State Space |
| | | | | **HW7 Due Mon 10/12**: Estimation |
| | Thu 10/15 | — | *Fall Break — No Class* | |

*Continued on next page*

| Wk | Date | Mtg | Type | Topic |
|---|---|---|---|---|
| 10 | Tue 10/20 | 17 | Content | Designing Control Systems for MIMO |
| | Thu 10/22 | 18 | Content | Native MIMO Controllers: LQR, LQG, MPC<br>**HW8 Due Mon 10/26**: MIMO Design |
| 11 | Tue 10/27 | 19 | Buffer | Integration Studio and Exam 2 Prep |
| | Thu 10/29 | 20 | Exam | **Exam 2**: MIMO, Robustness, Implementation |
| 12 | Tue 11/3 | 21 | Content | CNN and Transformer Encoders for State Estimation<br>**Project 2 Proposal Due Tue 11/3** |
| | Thu 11/5 | 22 | Content | Transformer World Models for Dynamics |
| 13 | Tue 11/10 | 23 | Content | Hybrid Model-Based and Learned Control<br>**HW9 Due Mon 11/9**: Learned Perception |
| | Thu 11/12 | 24 | Content | Differentiable Predictive Control |
| 14 | Tue 11/17 | 25 | Content | Diffusion Policies for Trajectory Tracking |
| | Thu 11/19 | 26 | Content | Generalization and Sim-to-Real<br>**HW10 Due Mon 11/23**: Learned Control |
| 15 | Tue 11/24 | — | *Thanksgiving Break — No Class* | |
| | Thu 11/26 | — | *Thanksgiving Break — No Class* | |
| 16 | Tue 12/1 | 27 | Presentation | Final Project Presentations Day 1 |
| | Thu 12/3 | 28 | Presentation | Final Project Presentations Day 2<br>**Project 2 Report Due Thu 12/3** |

## Assignment Summary

Each homework (HW1–HW10) consists of two files: $\mathbf{HW\_X_m}$ (math) and $\mathbf{HW\_X_c}$ (code).

## Important Dates

*This syllabus is subject to change. Students will be notified of any modifications.*

| **Assignment** | **Due Date** | **Topic Coverage** |
|---|---|---|
| HW0 | Sun 8/16 (midnight) | Mathematics Prerequisites |
| HW1 | Mon 8/24 | Modeling I (Meeting 1) |
| HW2 | Mon 8/31 | Modeling II: Actuators & Sensors (Meetings 2–3) |
| HW3 | Tue 9/8 | Basic Feedback Design (Meetings 4–5) |
| HW4 | Mon 9/14 | Stability Analysis (Meetings 6–7) |
| HW5 | Mon 9/21 | Architecture & State Space (Meetings 8–9) |
| HW6 | Mon 10/5 | Discrete & Nonlinear (Meetings 11–13) |
| HW7 | Mon 10/12 | State Estimation (Meeting 13) |
| HW8 | Mon 10/26 | MIMO Design (Meetings 16–18) |
| HW9 | Mon 11/9 | Learned Perception (Meetings 21–22) |
| HW10 | Mon 11/23 | Learned Control (Meetings 23–26) |
| **Exam 1** | Tue 9/22 | Modeling, Feedback, Stability |
| **Exam 2** | Thu 10/29 | MIMO, Robustness, Implementation |
| Project 1 Proposal | Thu 9/24 | — |
| Project 1 Report | Thu 10/8 | — |
| Project 2 Proposal | Tue 11/3 | — |
| Project 2 Report | Thu 12/3 | — |

Table 7: Complete assignment schedule with topic coverage.

| **Event** | **Date** |
|---|---|
| HW0 Due | Sunday, August 16, 2026 (midnight) |
| First Day of Class | Thursday, August 20, 2026 |
| Labor Day (University Closed) | Monday, September 7, 2026 |
| Exam 1 | Tuesday, September 22, 2026 |
| Project 1 Presentations | October 6–8, 2026 |
| Fall Break | Thursday–Friday, October 15–16, 2026 |
| Exam 2 | Thursday, October 29, 2026 |
| Thanksgiving Break | Tuesday–Thursday, November 24–26, 2026 |
| Project 2 Presentations | December 1–3, 2026 |
| Last Day of Class | Thursday, December 3, 2026 |
| Final Exam (Optional) | During Finals Week, December 7–11, 2026 |

Table 8: Key semester dates.